Companions hidden in public HARPS velocities: new stellar companions, a dark 1.2 $M_\odot$ companion to the giant HD 20037, and two planet candidates

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Abstract

HARPS-RVBank holds 212 562 public velocities of 2 912 stars, and Gaia DR3 measures the same stars' motions on the sky. Searched together, they reveal companions that no catalogue holds. A census of the velocities for variation beyond the errors, set against the NASA Exoplanet Archive, SB9 and Gaia DR3's two-body orbits, followed by Keplerian fits sampled with emcee and inclinations bounded by Gaia's astrometry, finds stellar and substellar companions of HD 107181, HD 77462, HD 7399 and HD 134985 with no published orbit. The G8 giant HD 20037, at 2.90 $M_\odot$, carries a companion whose orbit follows from its HARPS velocities, Gaia DR3's velocity-trend and acceleration solutions and its Hipparcos-Gaia proper-motion anomaly fitted together: a period of $28.5^{+3.2}_{-2.5}$ yr, an eccentricity of 0.46 ± 0.04, an inclination of $98.8 \pm 1.2°$ and a mass of $1.18 \pm 0.05\,M_\odot$. Its lines do not appear in the giant's spectra along its own track, where a main-sequence star of that mass, at 4 per cent of the light, would read at 7σ, so the companion is a massive white dwarf or a close pair of M dwarfs. A second census of the per-night velocities for periodic signals, with every star's activity indicators, its Ca II H and K emission measured in the HARPS spectra, its TESS photometry and a curved long-term trend and an offset across HARPS's 2015 fibre upgrade tested against each signal, leaves two planet candidates in no catalogue: HD 111564 b, $m \sin i = 0.29^{+0.11}_{-0.08}\,M_J$ at 856 d, and HD 222480 b, $23.6 \pm 2.3\,M_\oplus$ at 23.4 d.

Keywords: Techniques: radial velocities, Astrometry, Binaries: spectroscopic, White dwarfs, Planets and satellites: detection

1. Introduction

Radial-velocity surveys have watched thousands of nearby stars for two decades, and their archives hold more than their programmes published. HARPS-RVBank [1] recomputed every public HARPS velocity with SERVAL [2], corrected the instrument's nightly zero points, and attached each spectrum's activity indicators, so that a star's whole record can be read at once. Gaia DR3 adds two independent views of the same stars: its own spectroscopic velocities and their variability, with orbital and trend solutions for some, and astrometry whose acceleration and proper-motion anomaly against Hipparcos [3] bound a companion's mass and inclination [4].

This paper reads the two together for companions that no catalogue holds. Section 2 describes the data, Section 3 the stellar and substellar companions, Section 4 the dark companion of HD 20037 and Section 5 the planet candidates.

2. Data

The velocities are HARPS-RVBank's SERVAL velocities with their nightly zero points corrected, 2 912 stars and 212 562 velocities, with the chromatic index, differential line width, Hα and

Table 1: Companions with no published orbit: the velocities (nights), the orbit's period, semi-amplitude and eccentricity with their 16th to 84th percentiles, the companion's minimum mass for the primary's mass m_1, and Gaia's RUWE and the Hipparcos-Gaia proper-motion anomaly's signal to noise.

Star	Type	Velocities (nights)	P (d)	K ($\mathrm{km\,s^{-1}}$)	e	m_1 ($\mathrm{M_\odot}$)	$m_2 \sin i$ ($\mathrm{M_\odot}$)	RUWE, PMa S/N
HD 107181	G3 V	16 (11)	13.0	15.6 (15.0–16.2)	0.10 (0.06–0.14)	1.03	0.196 (0.187–0.204)	1.13, 38.9
HD 77462	F7 V	11 (11)	345 (338–355)	4.57 (4.37–4.85)	0.47 (0.45–0.49)	1.30	0.172 (0.165–0.183)	14.0, 5.8
HD 7399	F5 V	38 (22)	524 (493–558)	0.52 (0.51–0.55)	0.07 (0.04–0.11)	1.23	0.023 (0.022–0.025)	1.87, 15.1
HD 134985	K0 V	23 (23)	8700 (7000–11000)	0.59 (0.41–0.76)	0.37 (0.32–0.42)	0.90	0.050 (0.034–0.073)	1.17, 96.3

Na D indices and the pipeline's line width and bisector of each spectrum. Each star is matched to Gaia DR3 [4] as the brightest source within 30 arcsec of its J2000 position whose G lies within 2.5 mag of its V, which follows stars of high proper motion to Gaia's epoch, and to the Hipparcos-Gaia proper-motion anomaly of ref. [3] through its Hipparcos number. The public HARPS spectra of the stars discussed here come from the ESO Science Archive, their TESS light curves [5] from SPOC, TESS-SPOC or QLP through Lightkurve [6], and their ultraviolet and X-ray fluxes from GALEX [7], eROSITA's first survey [8], ROSAT's 2RXS [9] and 4XMM.

3. Stellar and substellar companions

A star enters the census when its velocities vary beyond their errors (a reduced χ^2 above 25, a scatter above $30\,\mathrm{m\,s^{-1}}$ and above five times the errors by the median absolute deviation, on four or more nights), when the SERVAL velocities agree with the pipeline's (correlation above 0.9), and when it is of spectral type F, G, K or M and luminosity class III or fainter. Of the 2 912 stars, 240 vary; 100 of them are explained by the NASA Exoplanet Archive, SB9 or a Gaia DR3 two-body orbit and 76 more are known binaries, variables or substellar hosts in SIMBAD. The 32 of the remaining 64 with ten or more nights were fitted with a Keplerian, and the thirteen strongest measured with a Keplerian, a linear trend and a jitter added to the errors, sampled with emcee [10], the periodogram's false-alarm level from shuffled velocities. Gaia bounds each orbit's inclination: for each of 200 orbits from the chain, 60 inclinations uniform in $\cos i$ and 24 nodes, the primary's photocentre, the companion taken dark, is followed through the Hipparcos and Gaia DR3 windows, its predicted proper-motion anomaly set against the observed one for periods beyond 300 d, and its predicted wobble over Gaia's window, at 0.3 mas a transit, against the RUWE. SIMBAD's references and the papers they cite give each star's literature.

Four companions have no published orbit (Table 1). HD 107181, a G3 dwarf, has a companion of at least $0.196\,\mathrm{M_\odot}$ on a 13.0 d orbit, and its velocities scatter by $1.2\,\mathrm{km\,s^{-1}}$ about that orbit while its proper-motion anomaly stands at 39σ, which the 13-day orbit cannot produce: a third body lies farther out. HD 77462, an F7 dwarf, has a companion of at least $0.172\,\mathrm{M_\odot}$ on an eccentric 345 d orbit, and its RUWE of 14.0 shows the star's wobble on the sky. HD 7399, an F5 dwarf, moves on a 524 d orbit at only $0.52\,\mathrm{km\,s^{-1}}$, a minimum mass of $24\,\mathrm{M_J}$ in the brown-dwarf desert; Gaia DR3 holds no orbit for it but an acceleration solution (Acceleration9, significance 26), and with its RUWE of 1.87 and proper-motion anomaly of 15σ the star moves more on the sky than $24\,\mathrm{M_J}$ at 524 d would move it, either because this companion is seen nearly face on, near $0.18\,\mathrm{M_\odot}$, or because a further body pulls the star. HD 134985 (GJ 579.3), a K0 dwarf, has a companion of at least $52\,\mathrm{M_J}$ on an orbit of about 24 yr, partly covered by its 8.3-year series, with a proper-motion anomaly of 96σ.

Table 2: HD 20037's orbit from its HARPS velocities and Gaia's and Hipparcos's astrometry together: medians with 16th to 84th percentiles of 4 000 samples.

Period (d)	10 423 (9 511–11 609)
Eccentricity	0.46 (0.43–0.51)
Giant's semi-amplitude K_1 ($\mathrm{km\,s^{-1}}$)	5.00 (4.95–5.11)
Inclination (deg)	98.8 (97.6–100.0)
Semi-major axis (AU)	14.9 (14.0–16.1)
Giant's mass ($\mathrm{M_\odot}$)	2.90 (2.76–3.07)
Companion's mass ($\mathrm{M_\odot}$)	1.18 (1.14–1.23)

4. The dark companion of HD 20037

HD 20037 is a G8 giant of $2.92 \pm 0.15\,\mathrm{M_\odot}$ and $10.3\,\mathrm{R_\odot}$ [3] at 172 pc. Its 13 HARPS velocities fall smoothly by $3.3\,\mathrm{km\,s^{-1}}$ from 2006 to 2011 with nights a day apart agreeing to a few metres per second. Gaia DR3 holds three more views of the same orbit. Its 27 velocities over 2014 to 2017 vary beyond their errors and are fitted by a second-degree trend (SecondDegreeTrendSB1) rising at $0.0071 \pm 0.0036\,\mathrm{km\,s^{-1}}$ per day at 2016.0; its astrometry carries an acceleration solution (Acceleration7) at a significance of 31.9, -1.468 ± 0.046 and $-0.705 \pm 0.047\,\mathrm{mas\,yr^{-2}}$ in right ascension and declination, with the proper motion at 2016.0; and its mean motion between Hipparcos and Gaia differs from that proper motion at 114σ [3].

One Keplerian orbit of the giant about the barycentre, the companion dark, is fitted to all of it at once. The HARPS velocities are read directly with a zero point and a jitter. Gaia's velocity derivatives are those of a quadratic fitted to the model's velocities at 27 times across its 968-day velocity window; the proper motion and acceleration at 2016.0 are those of a quadratic fitted to the photocentre over Gaia's astrometric window; and the long-term mean motion is the photocentre's displacement from its Hipparcos position, a line fitted over Hipparcos's window, to its Gaia position over the 24.75 yr between them. The parameters are the period, the time of periastron, $\sqrt{e}\cos\omega$, $\sqrt{e}\sin\omega$, the companion's mass, $\cos i$, the node, the HARPS zero point, the jitter and the giant's mass with its prior. The posterior was located by differential evolution from four starts and sampled with emcee. Every data set is met (Table 2): the proper-motion anomaly to within $0.03\,\mathrm{mas\,yr^{-1}}$, the acceleration to within 1σ, Gaia's velocity slope at $0.0080\,\mathrm{km\,s^{-1}}$ per day, and the HARPS velocities to $13\,\mathrm{m\,s^{-1}}$. The companion weighs $1.18 \pm 0.05\,\mathrm{M_\odot}$ on an orbit of 28.5 yr seen nearly edge on; periastron passed in 2013, between the HARPS and Gaia series.

A main-sequence star of $1.18\,\mathrm{M_\odot}$, an F8 or G0 dwarf, would give about 4 per cent of the system's V-band light. With the orbit known, the companion's lines can be sought along their own track: in each spectrum it moves against the giant at $-(1 + m_1/m_2)$ times the giant's velocity about the systemic one, here +5.0 to $+16.6\,\mathrm{km\,s^{-1}}$. The spectra are placed in the giant's frame on a grid uniform in velocity, divided by their median and whitened, cross-correlated with the line depths of a G0 dwarf, the rows' common pattern taken out, and read along that track. The track reads $z = -0.3$ against 2 000 shuffles of the rows' velocities ($p = 0.61$), while the same G0 spectrum laid into the actual spectra along the track at 0.5, 1, 2 and 4 per cent of the light reads $z = 1.6$, 3.7, 6.2 and 7.3. A single main-sequence companion of this mass is therefore excluded. GALEX measures HD 20037 at FUV 19.76 ± 0.11 and NUV 14.24 ± 0.01, a G8 giant's colours with no hot component. eROSITA records a soft source 1.8 arcsec from the star, 17 counts mostly between 0.5 and 1 keV, $2.0 \times 10^{29}\,\mathrm{erg\,s^{-1}}$ in the 0.2 to 2.3 keV band, as a giant's corona or an active pair of M

Table 3: The planet candidates: the host's type and distance, the nights and span of its HARPS series, and the orbit's period, semi-amplitude and eccentricity and the planet's minimum mass with their 16th to 84th percentiles; the false-alarm level of the host's calcium index near the period and its correlation with the velocities; and the Bayesian information criterion's gain of the Keplerian over a quadratic trend alone.

Planet	Host, distance	Nights (span, d)	P (d)	K ($\mathrm{m\,s^{-1}}$)	e	$m \sin i$	Ca II at P; r	BIC gain
HD 111564 b	G0 V, 51 pc	40 (2731)	856 (827–878)	6.6 (4.6–10.0)	0.39 (0.15–0.60)	0.29 (0.21–0.40) $\mathrm{M_J}$	1.00; 0.00	20.5
HD 222480 b	G1 V, 44 pc	21 (731)	23.41 (23.34–23.49)	5.3 (4.8–5.8)	0.21 (0.10–0.31)	23.6 (21.3–25.8) $\mathrm{M_\oplus}$	1.00; −0.35	–

dwarfs would give and far below a neutron star accreting the giant's wind.

The companion is therefore dark at 1.18 $\mathrm{M_\odot}$: a white dwarf in the upper few per cent of the mass distribution, whose progenitor was more massive than the giant is now, or a close binary of two M dwarfs near 0.6 $\mathrm{M_\odot}$ each, whose combined light, below 0.2 per cent of the giant's, the spectra cannot reach. The eROSITA source fits the second, but a G8 giant can supply it alone.

5. Planet candidates

The second census searches each star's velocities for periodic signals. The velocities and activity indicators are averaged per night, the velocity weighted by its inverse variance, and each F, G, K or M star of luminosity class III or fainter with 15 or more nights over 100 d or more, 1 145 stars, is searched by a generalised Lomb-Scargle periodogram from 1.5 d to twice its span, a signal standing at a false-alarm level [11] below 10^{-3}. A signal that an activity indicator shares within 5 per cent at a false-alarm level below 0.01, or with which one correlates at $|r| > 0.5$ and $p < 10^{-3}$, is set aside, as are the hosts of the NASA Exoplanet Archive, matched by host, HD or Hipparcos name or by position carried back to J2000, and the stars the first census explained. A Keplerian fitted from the three best periods leaves 108 signals below 85 Jupiter masses.

Each of these was set against Gaia DR3, the proper-motion anomaly, SB9, the Washington Double Star catalogue, the Open Exoplanet Catalogue, arXiv by every name of the star and SIMBAD's references, which return the known planets and companions. The strongest were fitted again with a Keplerian, a linear trend and a jitter sampled with emcee from 1.5 d upward, so that a one-day alias cannot stand in for a long period; the long periods were fitted again beside a quadratic trend, since a stellar companion's curvature can mimic a partial orbit, and with a free offset across HARPS's fibre upgrade of 2015 June where a series spans it; each star's Ca II H and K index was measured in every public HARPS spectrum and searched for the signal's period and correlated with the velocities; and each short period's TESS light curve was searched for rotation at the period or half of it and for transits.

Two candidates stand every test and appear in no catalogue or paper (Table 3), each observed on one side of the fibre upgrade. HD 111564, a G0 dwarf at 51 pc, carries a Saturn-mass candidate at 856 d, the highest peak of its residuals once a quadratic is taken out, with its calcium index flat at the period ($r = 0.00$). HD 222480, a G1 dwarf at 44 pc, carries a Neptune-mass candidate at 23.4 d beside a slow fall of 1.4 $\mathrm{m\,s^{-1}}$ a year; its calcium index is flat at the period ($r = -0.35$) and TESS shows it varying by 80 ppm with no period near 23 d.

The other signals fall to the same tests. HD 55693's 2361 d signal follows its calcium index, which peaks at 2280 d at a false-alarm level of 10^{-4} and correlates with the velocities at $r = 0.65$: the magnetic cycle found by ref. [13]. GJ 3102, HIP 92451 and HD 6790, Neptune-mass signals at 16.1, 16.0 and 12.4 d, vary in TESS near their periods or half of them, HIP 92451 at 7.5 to 8.6 d in seven of nine sectors, and GJ 3102's rotation is estimated near 13 d [14]; they are the stars' rotation

until shown otherwise. GJ 2049's 874 d signal runs to an eccentricity of 0.95 beside a quadratic. HD 129814, the solar twin whose velocities curve under a companion above $0.5\,M_\odot$ [12], carries a signal of 1120 d and $14.6\,m\,s^{-1}$ beside a linear trend, but its series spans the fibre upgrade with six of its 34 velocities after it, and with a free offset across the upgrade the orbit falls to an eccentricity of 0.83 to 0.95 at 500 to 600 d, so the signal is not separated from the instrument. None of the four short periods shows a transit that repeats in TESS.

6. Conclusion

Public HARPS velocities read beside Gaia DR3's astrometry and velocities give four stellar and substellar companions with no published orbit, among them a 13-day M dwarf in a triple and a brown-dwarf-desert minimum mass that the astrometry turns into a low-mass star. They give the giant HD 20037 a dark companion of $1.18 \pm 0.05\,M_\odot$ on a 28.5-year orbit, whose lines are absent where a main-sequence star of its mass would stand at 7σ: a massive white dwarf or a pair of M dwarfs. And after the activity, rotation, curvature and instrumental offsets that most periodic signals in the archive turn out to be, they leave two planet candidates: a Saturn-mass planet around HD 111564 and a Neptune-mass planet around HD 222480.

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References

[1] T. Trifonov et al., Public HARPS radial velocity database corrected for systematic errors, *Astron. Astrophys.* **636**, A74 (2020).

[2] M. Zechmeister et al., Spectrum radial velocity analyser (SERVAL). High-precision radial velocities and two alternative spectral indicators, *Astron. Astrophys.* **609**, A12 (2018).

[3] P. Kervella, F. Arenou and F. Thévenin, Stellar and substellar companions from Gaia EDR3. Proper-motion anomaly and resolved common proper-motion pairs, *Astron. Astrophys.* **657**, A7 (2022).

[4] Gaia Collaboration, F. Arenou et al., Gaia Data Release 3. Stellar multiplicity, a teaser for the hidden treasure, *Astron. Astrophys.* **674**, A34 (2023).

[5] G. R. Ricker et al., Transiting Exoplanet Survey Satellite, *J. Astron. Telesc. Instrum. Syst.* **1**, 014003 (2015).

[6] Lightkurve Collaboration, Lightkurve: Kepler and TESS time series analysis in Python, Astrophysics Source Code Library, ascl:1812.013 (2018).

[7] L. Bianchi, B. Shiao and D. Thilker, Revised catalog of GALEX ultraviolet sources. I. The all-sky survey: GUVcat_AIS, *Astrophys. J. Suppl.* **230**, 24 (2017).

[8] A. Merloni et al., The SRG/eROSITA all-sky survey. First X-ray catalogues and data release of the western Galactic hemisphere, *Astron. Astrophys.* **682**, A34 (2024).

[9] Th. Boller et al., Second ROSAT all-sky survey (2RXS) source catalogue, *Astron. Astrophys.* **588**, A103 (2016).

[10] D. Foreman-Mackey, D. W. Hogg, D. Lang and J. Goodman, emcee: the MCMC hammer, *Publ. Astron. Soc. Pac.* **125**, 306 (2013).

[11] R. V. Baluev, Assessing the statistical significance of periodogram peaks, *Mon. Not. R. Astron. Soc.* **385**, 1279 (2008).

[12] L. A. dos Santos et al., Spectroscopic binaries in the Solar Twin Planet Search program: from substellar-mass to M dwarf companions, *Mon. Not. R. Astron. Soc.* **472**, 3425 (2017).

[13] P. A. Peña Rojas and J. S. Jenkins, EMPEROR I. Exoplanet MCMC parallel tempering for RV orbit retrieval, *Astron. Astrophys.* **704**, A323 (2025).

[14] L. Mignon et al., Radial velocity homogeneous analysis of M dwarfs observed with HARPS I. Exoplanet detection and candidates, *Astron. Astrophys.* **689**, A32 (2024).